

EX -1 Spring Data JPA - Quick Example

The screenshot displays a database IDE (likely SQL Developer) and an IDE (likely Eclipse) used for a Spring Data JPA example.

Database IDE (Top):

- Connections:** Shows a connection to 'LocalXe'.
- Worksheet:** Contains SQL code:

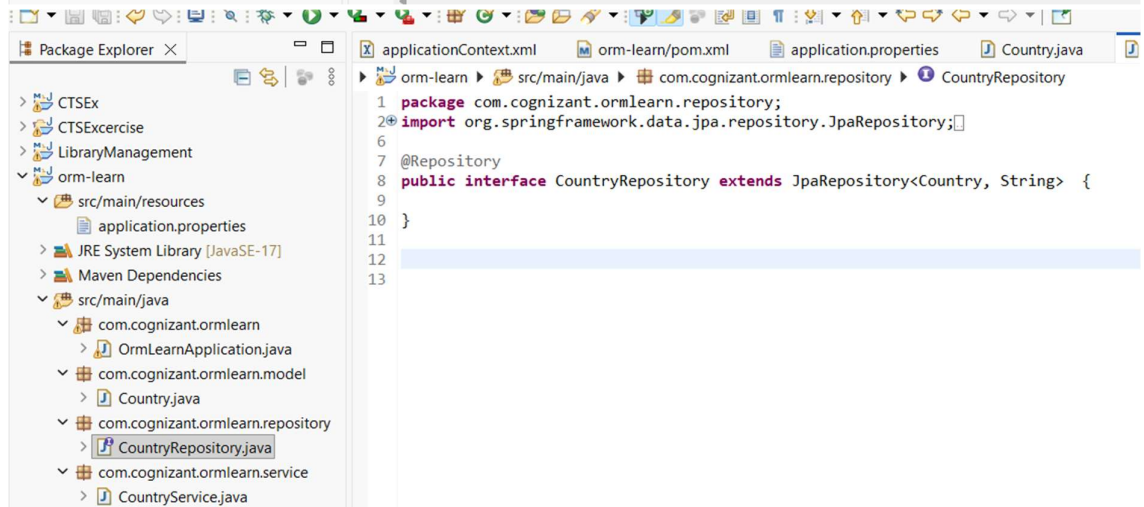
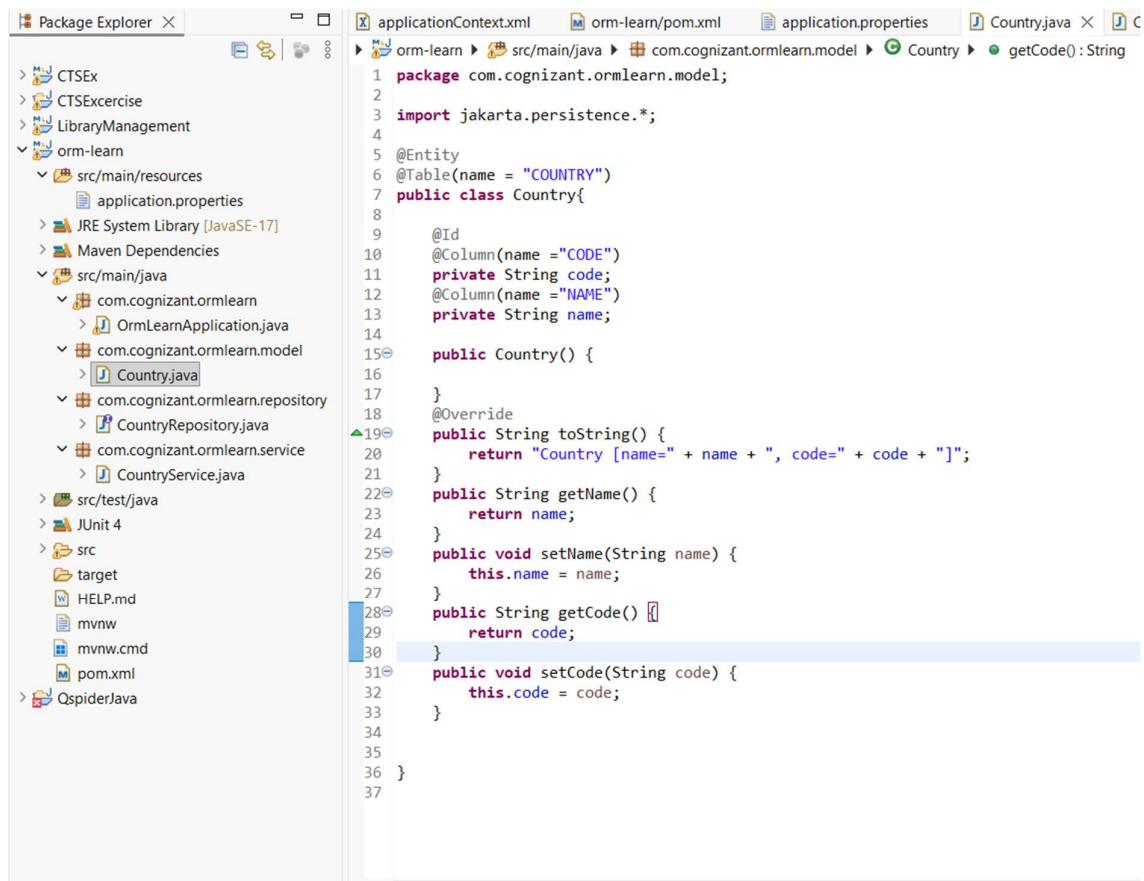
```
CREATE TABLE country (  
    code VARCHAR2(2) PRIMARY KEY,  
    name VARCHAR2(50)  
);  
  
INSERT INTO country VALUES ('IN', 'India');  
  
INSERT INTO country VALUES ('US', 'United States of America');  
  
COMMIT;  
  
SELECT * FROM country;  
  
SELECT USER FROM DUAL;
```
- Script Output:** Shows the execution results:

CODE	NAME
1 US	United States of America
2 IN	India

IDE (Bottom):

- Package Explorer:** Shows the project structure, including 'orm-learn' and 'CountryService.java'.
- applicationContext.xml:** Contains the following configuration:

```
1 logging.level.org.springframework=INFO  
2 logging.level.com.cognizant=DEBUG  
3  
4 spring.datasource.url=jdbc:oracle:thin:@localhost:1521/x  
5 spring.datasource.username=system  
6 spring.datasource.password=Hema@110306  
7 spring.datasource.driver-class-name=oracle.jdbc.OracleDriver  
8  
9 spring.jpa.hibernate.ddl-auto=validate  
10 spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.OracleDialect  
11  
12 spring.jpa.show-sql=true
```



The image shows a screenshot of an IDE with two tabs open, displaying Java code. The left tab shows the `CountryService` class, and the right tab shows the `OrmLearnApplication` class. Both files are part of the `com.cognizant.ormlearn.service` package.

CountryService.java

```
1 package com.cognizant.ormlearn.service;
2
3 import java.util.List;
4
5 import org.springframework.beans.factory.annotation.Autowired;
6 import org.springframework.stereotype.*;
7
8 import com.cognizant.ormlearn.model.Country;
9 import com.cognizant.ormlearn.repository.CountryRepository;
10
11 @Service
12 public class CountryService {
13
14     @Autowired
15     private CountryRepository repository;
16
17     public List<Country> getAllCountries()
18     {
19         return repository.findAll();
20     }
21 }
22
23
24
```

OrmLearnApplication.java

```
1 package com.cognizant.ormlearn;
2 import org.springframework.context.ApplicationContext;
3 import java.util.List;
4
5 import org.junit.Test;
6 import org.slf4j.Logger;
7 import org.springframework.boot.SpringApplication;
8 import org.springframework.boot.autoconfigure.SpringBootApplication;
9
10 import com.cognizant.ormlearn.model.Country;
11 import com.cognizant.ormlearn.service.CountryService;
12 import lombok.extern.slf4j.Slf4j;
13
14 @Slf4j
15 @SpringBootApplication
16 public class OrmLearnApplication {
17
18     private static CountryService countryService;
19
20     private static void testGetAllCountries() {
21         log.info("Start");
22         List<Country> countries = countryService.getAllCountries();
23         log.debug("countries={}", countries);
24         log.info("End");
25     }
26
27
28
29     public static void main(String[] args) {
30         ApplicationContext context = SpringApplication.run(OrmLearnApplication.class, args);
31         countryService = context.getBean(CountryService.class);
32
33         testGetAllCountries();
34     }
35 }
36
37
38
```

```
Console X
<terminated> OrmLearnApplication [Java Application] C:\Users\hemap\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_25.0.1.v20251108-1451\jre\bin\javaw.exe (30 Jun 2026, 11:01:50am - 11:01:57am elapsed: 0:00)

2026-06-30T11:01:52.204+05:30 INFO 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : Starting OrmLearnApplication using Java 25.0.1 with PID 35460 (C:\
2026-06-30T11:01:52.206+05:30 DEBUG 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : Running with Spring Boot v3.5.3, Spring v6.2.8
2026-06-30T11:01:52.206+05:30 INFO 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-06-30T11:01:52.260+05:30 INFO 35460 --- [ restartedMain] o.s.b.devtools.restart.ChangeableUrls : The Class-Path manifest attribute in C:\Users\hemap\m2\repository
2026-06-30T11:01:52.261+05:30 INFO 35460 --- [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-proper
2026-06-30T11:01:52.866+05:30 INFO 35460 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-06-30T11:01:52.938+05:30 INFO 35460 --- [ restartedMain] .s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 61 ms. Found 1 JPA rep
2026-06-30T11:01:53.489+05:30 INFO 35460 --- [ restartedMain] o.hibernate.jpa.internal.util.LogHelper : HHH000204: Processing PersistenceUnitInfo [name: default]
2026-06-30T11:01:53.564+05:30 INFO 35460 --- [ restartedMain] org.hibernate.Version : HHH000412: Hibernate ORM core version 6.6.18.Final
2026-06-30T11:01:53.600+05:30 INFO 35460 --- [ restartedMain] o.h.c.internal.RegionFactoryInitiator : HHH000026: Second-level cache disabled
2026-06-30T11:01:53.973+05:30 INFO 35460 --- [ restartedMain] o.s.o.j.p.SpringPersistenceUnitInfo : No LoadTimeWeaver setup: ignoring JPA class transformer
2026-06-30T11:01:54.011+05:30 INFO 35460 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-06-30T11:01:54.471+05:30 INFO 35460 --- [ restartedMain] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection oracle.jdbc.driver.T4CConnection@5
2026-06-30T11:01:54.472+05:30 INFO 35460 --- [ restartedMain] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-06-30T11:01:54.627+05:30 WARN 35460 --- [ restartedMain] org.hibernate.orm.deprecation : HHH90000025: OracleDialect does not need to be specified explicitl
2026-06-30T11:01:54.754+05:30 INFO 35460 --- [ restartedMain] org.hibernate.orm.connections.pooling : HHH10001005: Database info:
Database JDBC URL [Connecting through datasource 'HikariDataSource (HikariPool-1)']
Database driver: undefined/unknown
Database version: 21.3
Autocommit mode: undefined/unknown
Isolation level: undefined/unknown
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-06-30T11:01:55.612+05:30 INFO 35460 --- [ restartedMain] o.h.e.t.j.p.i.JtaPlatformInitiator : HHH000489: No JTA platform available (set 'hibernate.transaction.j
2026-06-30T11:01:56.277+05:30 INFO 35460 --- [ restartedMain] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-06-30T11:01:56.675+05:30 INFO 35460 --- [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-06-30T11:01:56.698+05:30 INFO 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : Started OrmLearnApplication in 4.946 seconds (process running for
2026-06-30T11:01:56.702+05:30 INFO 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : Start
Hibernate: select c1_0.code,c1_0.name from country c1_0
2026-06-30T11:01:56.984+05:30 DEBUG 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : countries=[Country [name=United States of America, code=US], Count
2026-06-30T11:01:56.984+05:30 INFO 35460 --- [ restartedMain] c.c.ormLearn.OrmLearnApplication : End
2026-06-30T11:01:56.996+05:30 INFO 35460 --- [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'default'
2026-06-30T11:01:57.002+05:30 INFO 35460 --- [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-06-30T11:01:57.027+05:30 INFO 35460 --- [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
```

EX-4 Difference between JPA, Hibernate and Spring Data JPA

1. Java Persistence API (JPA)

JPA (Java Persistence API) is a specification (set of rules) for storing, retrieving, updating, and deleting Java objects in a database.

It is defined by the Java specification (JSR 338).

JPA does not provide any implementation. It only defines interfaces and standards.

To use JPA, we need an implementation such as Hibernate.

Example: @Entity, @Table, @Id are JPA annotations.

2. Hibernate

Hibernate is an ORM (Object Relational Mapping) framework.

It is one of the most popular implementations of JPA.

Hibernate converts Java objects into database tables and SQL queries.

It provides features like session management, transaction management, caching, and query execution.

Here, the developer manually manages: Session ,Transaction ,Commit and Close

3. Spring Data JPA

Spring Data JPA is a Spring module built on top of JPA.

It does not implement JPA itself. Instead, it works with a JPA implementation like Hibernate.

It reduces boilerplate code by providing ready-made repository interfaces such as JpaRepository.

It automatically handles many database operations and transaction management.